Cumulative frequency tables and histograms

1 Consider the following table:

Score (x)	Cumulative frequency (cf)
10	7
11	15
12	18
13	20
14	26

a Find the total number of scores recorded.

b Find the number of times a score of 14 occured.

c Find the number of times a score less than 13 occured.

2 Complete the following table:

Score (x)	Frequency (f)	Cumulative frequency (cf)
18	10	
19	9	
20	3	22
21		
Total:	29	

3 The number of sightings of the Northern Lights were recorded across various Canadian locations over a period of 1 month. The numbers below represent the number of sightings at each location:

$$12, 8, 9, 8, 11, 7, 7, 11, 10, 9, 9, 11, 7, 10, 11, 7, 8, 9, 11, 9$$

a Construct a cumulative frequency table for this data.

b Find the number of locations where there were at least 11 sightings.

c Find the number of locations where there were less than 11 sightings.

d Find the median number of sightings across all 20 locations.

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4 A pair of dice are rolled 50 times and the numbers appearing on the uppermost face are added to give a score. The results are recorded in the given table:

Score (x)	Frequency (f)	Cumulative frequency (cf)
2	1	
3	2	
4	5	
5	5	
6	5	
7	9	
8	7	
9	5	
10	8	
11	1	
12	2	

a State the lowest possible score when a single pair of dice are rolled.

b State the highest possible score when a single pair of dice are rolled.

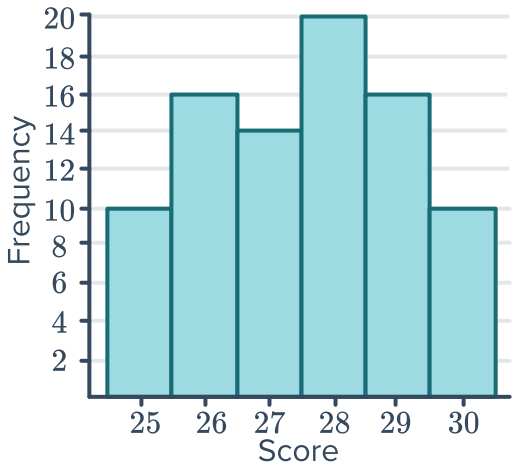
c Complete the table by finding the cumulative frequency values.

d Find the number of times a score of 8 occured.

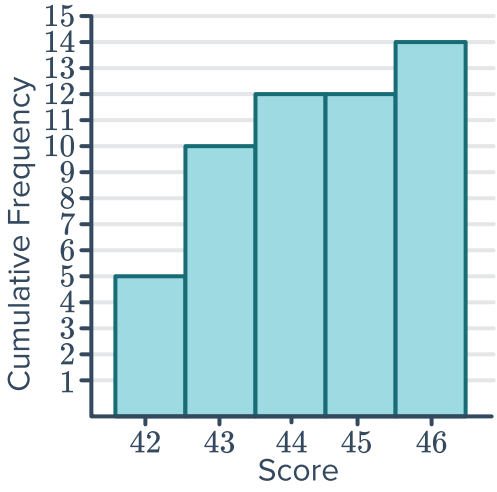
e Find the number of times a score more than 9 occured.

f Find the number of times a score of at most 6 occured.

5 Construct a cumulative frequency table for the data represented in the given histogram:



6 Consider the following cumulative frequency histogram:



a Find the total number of scores recorded.

b Find the number of times a score of 46 occured.

c Find the number of times a score of 45 occured.

d Find the percentage of scores that were 43 or less. Round your answer to one decimal place.

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7 Consider the frequency table showing the number of 'holes in one' across golf tournaments:

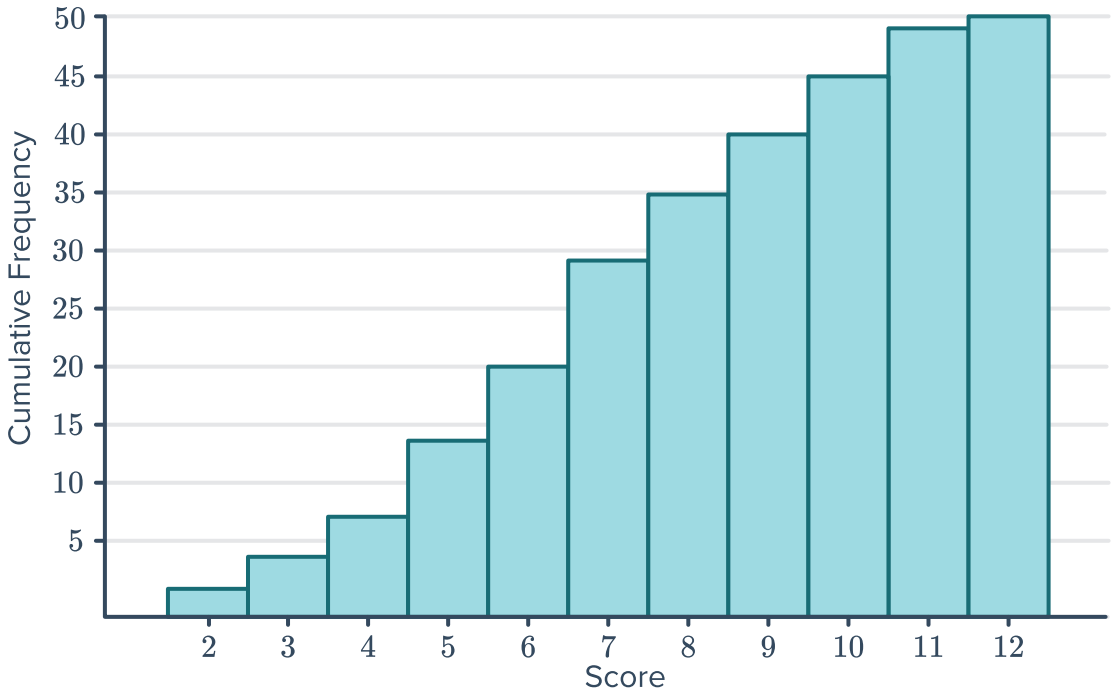
Number of 'holes in one'	Tournaments
2	5
3	1
4	3
5	4
6	0

a Construct cumulative frequency histogram for this data.

b Find the total number of 'holes in one' across all the tournaments.

c In how many tournaments were at most 3 'holes in one' scored?

8 A pair of dice are rolled 50 times and the numbers appearing on the uppermost face are added to give a score. The results are presented in the following histogram:



a Explain how the frequency of a certain score can be determined from a cumulative frequency histogram.

b What was the most frequent score?

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9 The marks received by a group of 20 students are given below:

$$75, 75, 95, 60, 95, 75, 60, 75, 95, 70, 70, 70, 60, 83, 75, 83, 70, 60, 75, 70$$

a Construct a cumulative frequency histogram for this data.

b Find the number of students who scored above 70.

c Find the number of students who scored less than or equal to 70.

d Find the number of students who scored above 80 but less than or equal to 90.

e Find the mode.

f Find the mean of the scores, correct to two decimal places.

g Find the median of the scores.

10 Families were asked how many times they got the flu during winter. The information has been partially filled out in the following table:

a Complete the frequency and cumulative frequency values in the given table.

b Find the total number of families that responded.

c Find the median number of times a family got the flu.

d Find the mean number of times a family got the flu. Round your answer to two decimal places.

e Construct a cumulative frequency histogram for the data.

f Find the number of families who got the flu more than twice.

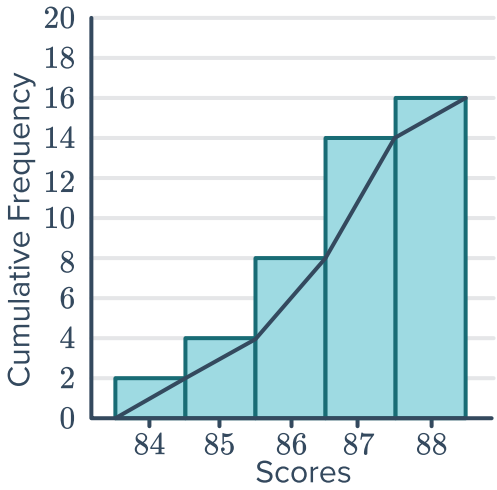
Score (x)	Frequency (f)	fx	Cumulative frequency
0	0	0	
1		2	
2		6	
3		6	
4		8	

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Cumulative frequency ogive

11 Use the cumulative frequency histogram to complete the frequency table:

Score	Frequency
84	
85	
86	
87	
88	



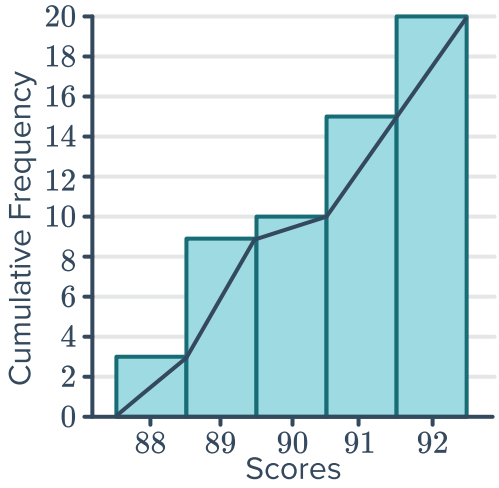
12 Consider the following cumulative frequency histogram and ogive:

a Use the ogive to determine the median score.

b Find the 14th lowest score.

c How many scores of 92 or less were there?

d How many scores fewer than 90 were there?



13 Consider the following cumulative frequency histogram and ogive:

Estimate:

a The 60th percentile.

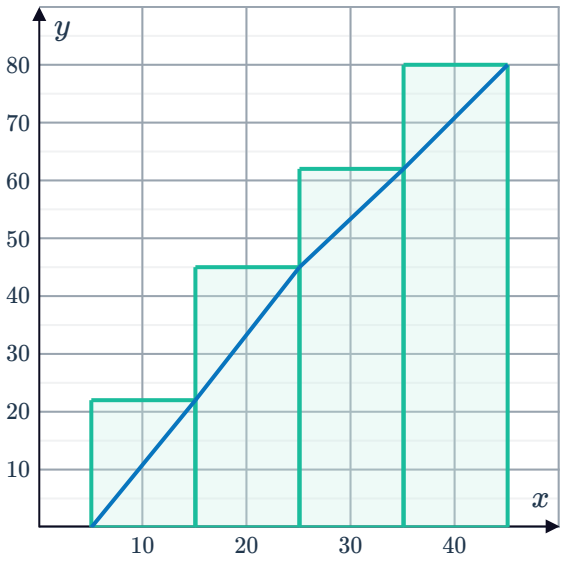
b The 10th percentile.

c The lower quartile.

d The median

e The upper quartile.

f The interquartile range.



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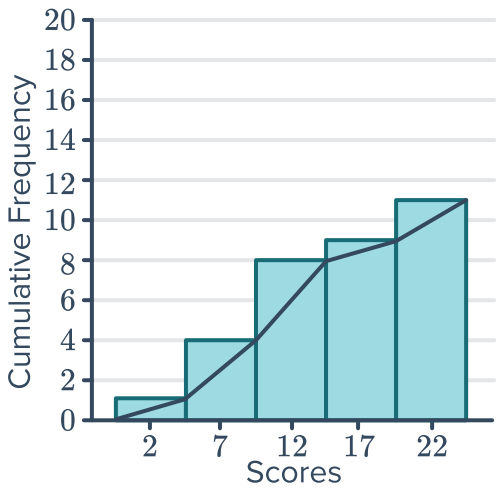
14 For each of the following graphs use the ogive to estimate:

i The median score.

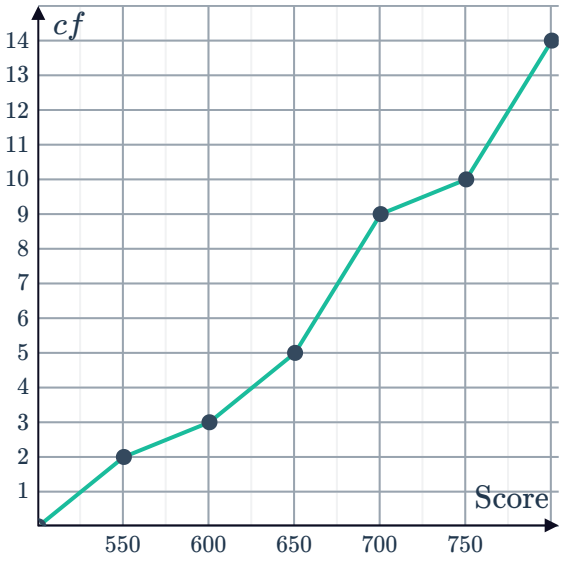
ii The lower quartile.

iii The upper quartile.

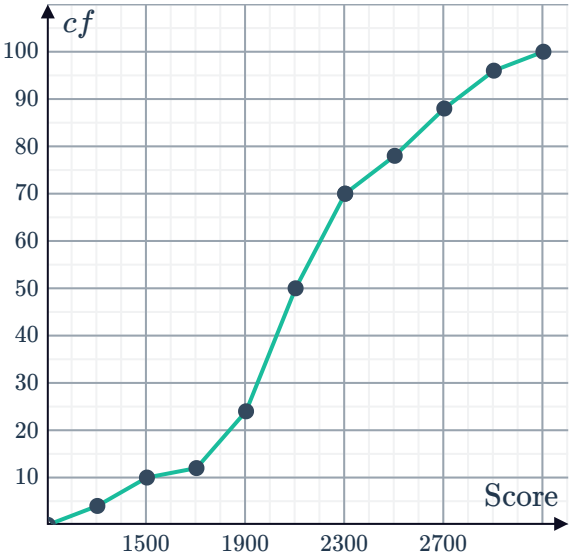
a



b



c



15 Consider the following frequency table:

a Complete the table by finding the cumulative frequency values.

b Construct a cumulative frequency histogram and ogive for the data.

c Find the median.

d Use your graph to estimate the 20th percentile.

Score (x)	Frequency (f)	Cumulative frequency (cf)
134	3	
135	2	
136	4	
137	6	
138	4	
Total:	19	

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Cumulative frequency and grouped data

16 For each of the frequency distribution tables below:

- **i** Complete the table.
- **ii** State the modal class.
- **iii** In which class interval does the median lie?
- **iv** Using the class centres, estimate the mean to two decimal places.

a

Class	Class centre	Frequency	Cumulative frequency
$1 - 7$		11	
$8 - 14$		14	
$15 - 21$		10	
$22 - 28$		15	
$29 - 35$		24	
Total:			

b

Class	Class centre	Frequency	Cumulative frequency
$1 - 9$		8	
$10 - 18$		16	
$19 - 27$		4	
$28 - 36$		21	
$37 - 45$		16	
Total:			

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17 Consider the frequency distribution table below:

Class	Class centre	Frequency	Cumulative frequency
$1-9$		3	
$10-18$		4	
$19-27$		3	
$28-36$		3	
$37-45$		8	
Total:			

a Complete the table.

b Construct a cumulative frequency histogram for the data.

c State the modal class.

d In which class interval does the median lie?

18 Consider the frequency distribution table below:

Score (x)	Frequency (f)	Cumulative frequency (cf)
$1-4$	4	
$5-8$	7	
$9-12$	11	
$13-16$	7	
$17-20$	4	

a Complete the table.

b Calculate the total frequency.

c Identify the class size.

d Describe the shape of the distribution.

e Approximately one third of the scores recorded are greater than what score?

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19 Consider the frequency distribution table below:

Score (x)	Frequency (f)	Cumulative frequency (cf)
$1-5$	7	
$6-10$	15	
$11-15$	8	
$16-20$	11	
$21-25$	2	
$26-30$	1	

a Complete the table.

b Calculate the total frequency.

c Identify the class size.

d Describe the shape of the distribution.

e Approximately half of the scores recorded are greater than what score?

20 Consider the frequency distribution table below:

Score (x)	Frequency (f)	Cumulative frequency (cf)
$20-24$	9	
$25-29$	18	
$30-34$	37	
$35-39$	14	
$40-44$	9	
$45-49$	6	
$50-54$	3	

a Complete the table.

b Calculate the total frequency.

c Identify the class size.

d Describe the shape of the distribution.

e Approximately one third of the scores recorded are greater than what score?

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21 Consider the set of scores given below:

$$58, 59, 70, 64, 69, 73, 64, 68, 59, 60, 54, 73$$

a Complete the following frequency table:

Class interval	Class centre (cc)	Frequency (f)	$f \times cc$	Cumulative frequency
$51 - 55$				
$56 - 60$				
$61 - 65$				
$66 - 70$				
$71 - 75$				
Total				

b State the modal class.

c Using the class centres, estimate the mean to one decimal place.

22 A group of high school students wanted to convince their principal that the school needed air-conditioning. They measured the temperature, t, in a classroom at 1 pm every day during February and recorded the results (in $^\circ\text{C}$) below:

$$35, 23, 24, 23, 30, 29, 36, 24, 35, 32, 29, 36, 33, 30, 30,$$
$$26, 27, 24, 32, 30, 28, 27, 24, 31, 26, 36, 29, 35, 30$$

a Construct a cumulative frequency table for the data. Use the discrete intervals of $23 - 25$, $26 - 28$, and etc.

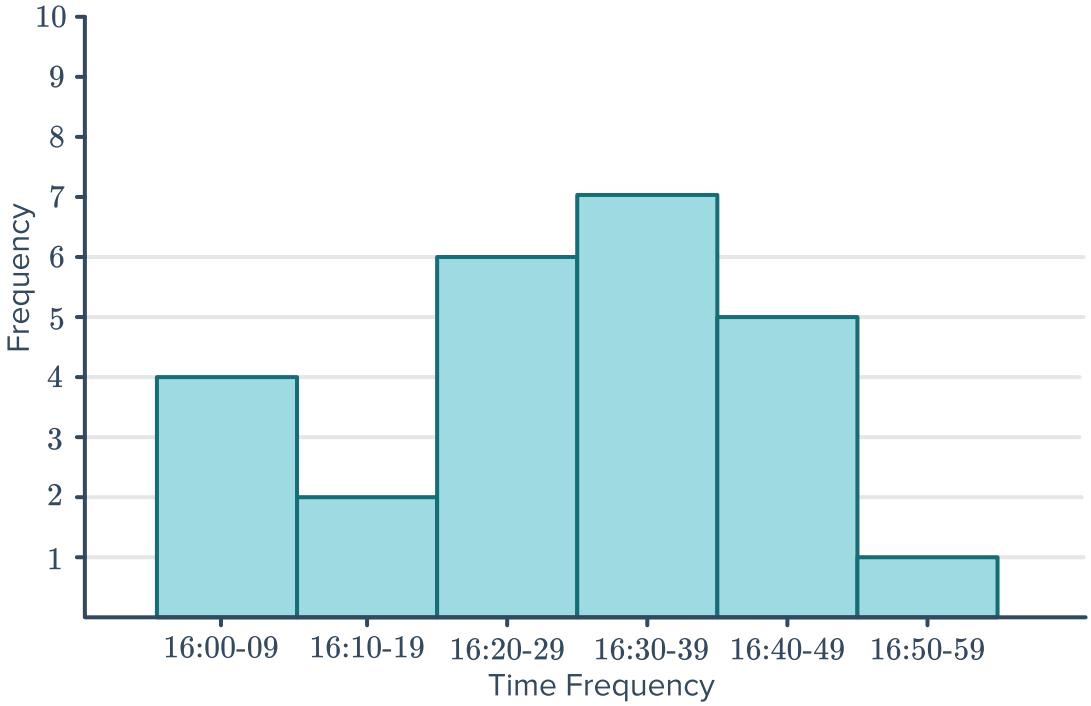
b Construct a cumulative frequency histogram for the data.

c Find the modal class.

d Using the class centres, estimate the average temperature to two decimal places.

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23 A $1500\ \text{m}$ swimmer records her time over several training sessions. Her times are recorded in the following histogram:



a Construct a cumulative frequency table for the data using the given intervals.

b Find the total number of training sessions she completed.

c In which class interval does the median time fall?

d Find the number of times she recorded a swim time faster than 16:50.

e Find the percentage of swims that were less than 16:30.

24 The heights of 22 boys in a class are listed below:

$$133, 150, 133, 142, 149, 146, 135, 150, 143, 140, 151,$$
$$151, 132, 149, 149, 138, 142, 149, 136, 147, 143, 145$$

a Construct a cumulative frequency histogram for the data. Use the discrete intervals of $131 - 135, 136 - 140$, etc.

b Find the number of students taller than $150\ \text{cm}$.

c Find the number of students who are at most $150\ \text{cm}$ tall.

d Find the number of students taller than $140\ \text{cm}$ but shorter than $146\ \text{cm}$.

e State the modal height class of the students.

f Calculate the centre of the class $146 - 150$.

g Which class contains the median height?

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25 A tree farmer measured the heights of some newly planted saplings to the nearest centimetre. The results were as follows:

$$20, 25, 25, 25, 24, 30, 25, 29, 24, 15, 24, 15, 34, 25, 30, 25,$$
$$14, 14, 35, 19, 25, 15, 15, 29, 35, 20, 25, 30, 25, 30, 24, 25, 24$$

a Find the range.

b Construct a cumulative frequency histogram for the data. Use the discrete intervals of $13 - 17, 18 - 22$, etc.

c Using the class centres, estimate the mean to two decimal places.

d In which class interval does the median lie?

e State the modal class.

f Find the percentage of saplings that are at least $23\ \text{cm}$ tall. Round your answer to one decimal place.

g Find the percentage of saplings that have a height lower than $23\ \text{cm}$. Round your answer to one decimal place.

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26 Scientists wanting to determine the effect of fatigue on reaction time while driving. They measured the reaction time, t, of several drivers at night. The results are presented in the following table:

Reaction time (seconds)	Class centre	Frequency	Cumulative frequency
$0.01 \leq t < 0.05$	0.03	29	29
$0.05 \leq t < 0.09$	0.07	32	61
$0.09 \leq t < 0.13$	0.11	39	100
$0.13 \leq t < 0.17$	0.15	31	131
$0.17 \leq t < 0.21$	0.19	36	167
$0.21 \leq t < 0.25$	0.23	33	200

a Using the class centres, calculate an estimate of:

i The median **ii** The mean

b One researcher wants to reduce the amount of data by increasing the size of each class interval. Complete the table:

Reaction Time (seconds)	Class Centre	Frequency	Cumulative frequency
$0.01 \leq t < 0.09$			
$0.09 \leq t < 0.17$			
$0.17 \leq t < 0.25$			

c Using the new class centres, calculate an estimate of:

i The median **ii** The mean

d By increasing the size of each class interval, by what percentage has the mean changed?

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